

SHIVANG RAI

Full-Stack Developer · AI Automation & Data Science

[Gmail](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#) | +91 7905192935 | Lucknow, India

SUMMARY

Full-Stack Developer with experience building scalable web applications, AI-powered systems, and data-driven platforms. Skilled in React.js, Node.js, Python, FastAPI, PostgreSQL, cloud technologies, AI agent orchestration, automation workflows, real-time systems, data analytics, and DevOps practices. Focused on delivering secure, high-performance, and production-ready solutions.

TECHNICAL SKILLS

Frontend: React.js, Redux, React Router, Tailwind CSS, Vite, Next.js

Backend: Node.js, Express.js, Fastify, TypeScript, JavaScript, Python, FastAPI, Django

AI & Automation: AI Agents (multi-agent pipelines), Gemini API, MCP (Model Context Protocol), Pydantic-validated LLM outputs, automation workflows

Data & Analytics: Pandas, NumPy, Data Visualization, Data Cleaning, EDA

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis, Firebase Firestore

DevOps & Cloud: Docker, GitHub Actions, Nginx, AWS (EC2, S3, Lambda), Firebase, Vercel, Render

Languages: Python, TypeScript, JavaScript, Java, C++

PROJECTS

TRAFFIC INTELLIGENCE SYSTEM

- Built a full-stack traffic analytics platform using React, Python, FastAPI, and pandas to analyze road accident and transport datasets.
- Developed interactive dashboards and visualizations for traffic insights and decision-making.
- Designed REST APIs and automated data-processing pipelines.

REAL-TIME CHAT APPLICATION

- Built a real-time chat application using React.js, Firebase, and WebRTC.
- Implemented one-to-one/group chats, encrypted messaging, media sharing, status stories, and presence tracking.
- Integrated voice/video calling with scalable real-time architecture.

APTICO | CAREER PROGRESS & GAMIFICATION PLATFORM

- Architected a 3-agent AI pipeline (Auditor, Strategist, Copywriter) that automates ad campaign performance analysis and copy rewriting, with each agent's output feeding the next to keep logic decoupled and testable.
- Implemented real-time Server-Sent Events streaming from FastAPI so users can watch each agent's output live in the React dashboard with zero polling; enforced Pydantic schemas to guarantee structured JSON from the Gemini API, eliminating untyped output failures

THE ROASTING HOUSE

- Designed REST APIs for product, cart, and order lifecycle management.
- Integrated cloud storage uploads and CI/CD deployment workflows.
- Applied TypeScript strict mode and production-grade backend practices.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Information Technology *Expected Graduation: May 2027*

Babu Banarasi Das University

Relevant Coursework: DSA, DBMS, Operating Systems, Computer Networks